1. Grasshopper plugins

Plugin	Purpose
Ladybug Tools (Ladybug + Honeybee)	Daylight, solar radiation, energy loads
Butterfly	CFD / airflow simulation
Karamba3D	Structural model components used in the definition
GhPython	Python 3 Script components (usually pre-installed with Rhino 8)

2. External simulation engines

Engine	Used by	Notes
blueCFD-Core 2017 (OpenFOAM-5.x)	Butterfly	**Default / recommended.** Native support for Butterfly's ABL boundary conditions.
Radiance	Honeybee	Daylight recipes (Honeybee can auto-install it)
EnergyPlus	Honeybee	Annual energy loads (Honeybee can auto-install it)

3. Optional / networking

- **Microsoft MPI** — included with blueCFD-Core; no separate install needed.
- **EPW weather file** — select a climate file for the site location placed in the project folder.

Note if you use blueCFD-Core 2020 instead

The script also works with **blueCFD-Core 2020 / OpenFOAM-8**, but the following changes apply:

- The embedded compatibility patch **auto-detects** OpenFOAM-8 and replaces the AML boundary conditions with standard `fixedValue` inlets and `nutkWallFunction`.
- The inlet becomes a **uniform velocity profile** instead of a height-dependent atmospheric boundary layer profile.
- Wind velocity results near the ground and around buildings may differ from the blueCFD 2017 version.

- Make sure Butterfly's environment scripts (`runmanagerenv.py`, `runmanager_bluecfd.py`) and the `setvars.bat` fallback are configured for OpenFOAM-8.